Production-Constrained Spatial Interaction Model

for Trip Distribution Estimation under Data Scarcity

Point Estimate Formulation (MLE / MAP) • Implementation Guide

Overview. This document presents a point-estimate formulation of the production-constrained spatial interaction model for estimating origin-destination (OD) trip flows when direct OD survey data are unavailable. Rather than computing a full posterior distribution, parameters are estimated by maximizing a penalized log-likelihood whose two components are (i) a multinomial distance-bin likelihood that anchors the deterrence parameters to observed aggregate distance proportions, and (ii) an L2 penalty on the destination attractiveness coefficients derived from OpenStreetMap (OSM) covariates. The resulting optimization problem is smooth and can be solved efficiently with standard gradient-based methods. Bootstrap resampling is used to quantify estimation uncertainty.

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1 Notation and Data Inputs

Let the study area be partitioned into N origin zones indexed by $i \in \{1, \ldots, N\}$ and M destination zones indexed by $j \in \{1, \ldots, M\}$. Table 1 lists all quantities that are observed or computable prior to estimation.

Table 1: Model notation and data sources.

Symbol	**Description**	**Source / Notes**
O_i	Total trips generated at origin i	Observed (household survey, census, or mobile data)
d_{ij}	Network distance from zone i to zone j	Computed via road network (e.g. OSMnx)
$\mathbf{x}_j \in \mathbb{R}^P$	OSM covariate vector for destination zone j	POI density, road centrality, land use, etc.
b_k	Observed proportion of trips in distance bin k	Empirical input; $\sum_k b_k = 1$
$[l_k, u_k)$	Lower and upper bounds of distance bin k	e.g. $[0, 3)$, $[3, 6)$, $[6, \infty)$ km
K	Number of distance bins	e.g. $K = 3$
$n = \sum_i O_i$	Total observed trips across all origins	Used to scale the multinomial likelihood
$\hat{T}_{ij}$	Estimated trip flow from i to j	Model output

2 Model Structure

2.1 Deterrence Function

Trip flows decrease with distance according to the **Tanner (combined) deterrence function**:

$$f(d_{ij};\, \beta, \gamma) \;=\; d_{ij}^{-\gamma} \cdot \exp(-\beta\, d_{ij})\,, \qquad \beta > 0, \quad \gamma \geq 0. \tag{1}$$

Setting $\gamma = 0$ recovers the pure exponential form; setting $\beta \to 0$ recovers the pure power-law. Both nested forms should be compared via AIC/BIC during model selection.

2.2 Destination Attractiveness

The attractiveness of destination j is modelled as a log-linear function of its OSM covariates:

$$A_j(\boldsymbol{\alpha}) \;=\; \exp\!\big(\boldsymbol{\alpha}^\top \mathbf{x}_j\big)\,, \qquad \boldsymbol{\alpha} \in \mathbb{R}^P. \tag{2}$$

2.3 Expected Trip Flow (Production Constraint)

The expected flow from origin i to destination j is:

$$\hat{T}_{ij} \;=\; O_i \cdot \underbrace{\frac{A_j(\boldsymbol{\alpha}) \cdot f(d_{ij};\, \beta, \gamma)}{\sum_{j'=1}^{M} A_{j'}(\boldsymbol{\alpha}) \cdot f(d_{ij'};\, \beta, \gamma)}}_{=\, B_i^{-1}(\boldsymbol{\alpha},\beta,\gamma)\ \text{(balancing factor)}}. \tag{3}$$

The denominator is the **origin-specific balancing factor** B_i^{-1}, which enforces the production constraint $\sum_j \hat{T}_{ij} = O_i$ exactly for every origin i.

Remark. *Compute* $\log B_i^{-1} = \text{log_sum_exp}_{j'}\left[\boldsymbol{\alpha}^\top \mathbf{x}_{j'} + \log f(d_{ij'})\right]$ *for numerical stability.*

3 Identification via the Distance Bin Likelihood

Because no individual OD counts are observed, the parameters $(\beta, \gamma, \boldsymbol{\alpha})$ are identified through the **aggregate distance distribution**. Define the model-implied proportion of trips in bin k:

$$\hat{p}_k(\beta, \gamma, \boldsymbol{\alpha}) \;=\; \frac{\sum_i \sum_j \hat{T}_{ij} \cdot \mathbf{1}[d_{ij} \in [l_k, u_k)]}{n}. \tag{4}$$

The multinomial log-likelihood of the observed bin counts $\mathbf{b} = (b_1, \ldots, b_K)$ given $n = \sum_i O_i$ total trips is:

$$\ell_{\text{dist}}(\beta, \gamma, \boldsymbol{\alpha}) \;=\; \sum_{k=1}^{K} n\, b_k \cdot \log \hat{p}_k(\beta, \gamma, \boldsymbol{\alpha})\,. \tag{5}$$

This is the *sole identification source* when no OD data are available. It directly links all three parameter groups to observable aggregate behavior.

4 Penalized Objective Function

4.1 Full Objective

The parameters are estimated by maximizing the following **penalized log-likelihood**:

$$\hat{\Theta} \;=\; \underset{\beta>0,\, \gamma\geq 0,\, \boldsymbol{\alpha}}{\arg\max}\; \mathcal{Q}(\beta, \gamma, \boldsymbol{\alpha})\,, \tag{6}$$

where

$$\mathcal{Q}(\beta, \gamma, \boldsymbol{\alpha}) \;=\; \underbrace{\ell_{\text{dist}}(\beta, \gamma, \boldsymbol{\alpha})}_{\text{distance bin fit}} - \underbrace{\frac{\lambda}{2}\,\|\boldsymbol{\alpha}\|^2}_{\text{L2 regularization}} \quad . \tag{7}$$

The scalar $\lambda > 0$ is the regularization strength, which penalizes large covariate coefficients and prevents overfitting to the limited distance bin data.

Remark. *Equation* (7) *is the maximum a posteriori (MAP) objective that corresponds to the Bayesian model with a Gaussian prior* $\alpha_k \sim \mathcal{N}(0, \sigma_\alpha^2)$*, where* $\lambda = 1/\sigma_\alpha^2$*. If the goal is pure MLE (no regularization), set* $\lambda = 0$*; however, regularization is strongly recommended when* P *is large relative to* K*.*

4.2 Extension: Partial OD Data

If OD counts T_{ij} are observed for a subset of pairs $\mathcal{S} \subseteq \{(i,j)\}$, add a Poisson log-likelihood term:

$$\ell_{\text{OD}}(\beta, \gamma, \boldsymbol{\alpha}) \;=\; \sum_{(i,j)\,\in\,\mathcal{S}} \Big[T_{ij}\,\log \hat{T}_{ij} - \hat{T}_{ij} - \log T_{ij}! \Big], \tag{8}$$

and maximize the combined objective:

$$\mathcal{Q}_{\text{full}} \;=\; \ell_{\text{OD}} + \ell_{\text{dist}} - \frac{\lambda}{2}\,\|\boldsymbol{\alpha}\|^2. \tag{9}$$

5 Parameter Initialization

Good initialization is critical for convergence. The following **closed-form moment estimates** provide reliable starting values.

5.1 Initialization of β

Under pure exponential deterrence and uniform destination density, the mean trip distance satisfies $\mathbb{E}[d] = 1/\beta$. Estimate the empirical mean distance from the observed bins:

$$\bar{d}_{\text{obs}} \;=\; \sum_{k=1}^{K} b_k \cdot m_k, \tag{10}$$

where m_k is the midpoint of bin k. The initial value is:

$$\beta^{(0)} \;=\; \frac{1}{\bar{d}_{\text{obs}}}. \tag{11}$$

Worked example. For bins $[0,3)$, $[3,6)$, $[6,\infty)$ with proportions $\mathbf{b} = (0.20,\ 0.70,\ 0.10)$ and midpoints (1.5, 4.5, 7.5) km:

$$\bar{d}_{\text{obs}} \;=\; 0.20 \times 1.5 + 0.70 \times 4.5 + 0.10 \times 7.5 \;=\; 4.20 \text{ km}, \qquad \beta^{(0)} \;\approx\; 0.238 \text{ km}^{-1}. \tag{12}$$

5.2 Initialization of γ

Set $\gamma^{(0)} = 1.0$, the center of the empirically observed range [0.5, 2.0] for urban contexts (Simini et al., 2012; Lenormand et al., 2016).

5.3 Initialization of $\boldsymbol{\alpha}$

Set $\boldsymbol{\alpha}^{(0)} = \mathbf{0}$, corresponding to equal attractiveness across all destinations before any covariate information is used.

6 Optimization

6.1 Gradient Expressions

The objective $\mathcal{Q}$ is differentiable everywhere in the interior of the feasible region. Let $w_{ij} = \hat{T}_{ij}/O_i$ denote the estimated choice probability from origin i to destination j, and let $\delta_{ij}^{(k)} = \mathbf{1}[d_{ij} \in [l_k, u_k)]$. Define the residual weight for bin k as:

$$r_k \;=\; \frac{n\, b_k}{\hat{p}_k} \;-\; n, \tag{13}$$

which measures the discrepancy between observed and model-implied bin proportions. The gradients of ℓ_{dist} with respect to each parameter are then:

$$\frac{\partial \ell_{\text{dist}}}{\partial \beta} \;=\; \sum_k r_k \sum_i \sum_j w_{ij} \left[\delta_{ij}^{(k)} - \hat{p}_k\right] \cdot (-d_{ij}), \tag{14}$$

$$\frac{\partial \ell_{\text{dist}}}{\partial \gamma} \;=\; \sum_k r_k \sum_i \sum_j w_{ij} \left[\delta_{ij}^{(k)} - \hat{p}_k\right] \cdot (-\log d_{ij}), \tag{15}$$

$$\frac{\partial \ell_{\text{dist}}}{\partial \alpha_p} \;=\; \sum_k r_k \sum_i \sum_j w_{ij} \left[\delta_{ij}^{(k)} - \hat{p}_k\right] \cdot (x_{jp} - \bar{x}_{ip}), \tag{16}$$

where $\bar{x}_{ip} = \sum_{j'} w_{ij'}\, x_{j'p}$ is the attractiveness-weighted mean of feature p for origin i. The gradient of the full penalized objective adds the regularization term:

$$\frac{\partial \mathcal{Q}}{\partial \alpha_p} \;=\; \frac{\partial \ell_{\text{dist}}}{\partial \alpha_p} \;-\; \lambda\, \alpha_p. \tag{17}$$

6.2 Constraints and Reparameterization

The positivity constraints $\beta > 0$ and $\gamma \geq 0$ are handled by reparameterizing before passing to an unconstrained optimizer:

$$\beta = \exp(\tilde{\beta}), \qquad \gamma = \exp(\tilde{\gamma}), \tag{18}$$

and optimizing over $(\tilde{\beta}, \tilde{\gamma}, \boldsymbol{\alpha}) \in \mathbb{R}^{P+2}$ without constraints. Apply the chain rule to transform the gradients accordingly.

6.3 Recommended Algorithm

Use the L-BFGS-B algorithm (available in `scipy.optimize.minimize` with `method='L-BFGS-B'`). L-BFGS-B is well-suited here because the objective is smooth, the parameter dimension is moderate ($P + 2$ free parameters), and bounded constraints on $\tilde{\beta}$, $\tilde{\gamma}$ can optionally be enforced directly.

Algorithm 1 Point Estimate Optimization

```
Compute d̄_obs from Equation (10); set β^(0) = 1/d̄_obs, γ^(0) = 1, α^(0) = 0
Reparameterize: β̃^(0) = log β^(0), γ̃^(0) = log γ^(0)
Define θ = (β̃, γ̃, α)
repeat
    Compute T̂_ij via Equation (3) for all (i, j)
    Compute p̂_k via Equation (4) for all k
    Evaluate Q(θ) via Equation (7)
    Compute gradient ∇_θ Q via Equations (14), (15) and (17)
    Update θ via L-BFGS-B step
until ‖∇_θ Q‖ < ε or maximum iterations reached
Recover β̂ = exp(β̃*), γ̂ = exp(γ̃*), α̂
Compute T̂_ij for all pairs via Equation (3)
```

Remark. *Run the optimizer from at least 10 random restarts by perturbing the initialization with* $\tilde{\beta}^{(0)} \sim \mathcal{N}(\log \beta^{(0)}, 0.25)$, $\tilde{\gamma}^{(0)} \sim \mathcal{N}(0, 0.25)$, *and* $\boldsymbol{\alpha}^{(0)} \sim \mathcal{N}(\mathbf{0}, 0.1\,\mathbf{I})$. *Report the solution with the highest* $\mathcal{Q}$ *value. This guards against convergence to local optima.*

7 Regularization Parameter Selection

The regularization strength λ controls the trade-off between fitting the distance bins and shrinking the covariate coefficients toward zero. Select λ by **cross-validation on the distance bin fit**:

1. Construct $V = 5$ folds by randomly partitioning the origin zones $\{1, \ldots, N\}$ into held-out groups.

2. For each fold v, compute the distance bin proportions $\mathbf{b}^{(-v)}$ using only the origins *not* in fold v, and fit the model on $\mathbf{b}^{(-v)}$.
3. Evaluate the held-out multinomial log-likelihood on the trips originating from the held-out zones.
4. Select $\hat{\lambda} = \arg\max_\lambda \frac{1}{V}\sum_{v=1}^{V} \ell_{\text{dist}}^{(-v)}(\lambda)$.

A practical search grid is $\lambda \in \{0.001, 0.01, 0.1, 1, 10\}$.

Remark. *When the number of distance bins K is small (e.g. $K = 3$), the model has few effective degrees of freedom and the distance bin fit is not very sensitive to λ. In this case, a default value of $\lambda = 0.1$ is reasonable without cross-validation.*

8 Uncertainty Quantification via Bootstrap

Because we work with point estimates, uncertainty is quantified through **parametric bootstrap resampling** rather than MCMC posteriors.

8.1 Bootstrap Procedure

1. Compute the point estimate $\hat{\Theta} = (\hat{\beta}, \hat{\gamma}, \hat{\boldsymbol{\alpha}})$ and the fitted flows $\hat{T}_{ij}$.
2. For bootstrap replicate $b = 1, \ldots, B$ (recommend $B = 500$):
   (a) Draw a bootstrap distance bin vector: $\mathbf{b}^{(b)} \sim \text{Multinomial}(n,\, \hat{\mathbf{p}})\ /\ n$.
   (b) Re-fit the model on $\mathbf{b}^{(b)}$ from the same starting point $\hat{\Theta}$.
   (c) Store the resulting estimates $\hat{\Theta}^{(b)} = (\hat{\beta}^{(b)}, \hat{\gamma}^{(b)}, \hat{\boldsymbol{\alpha}}^{(b)})$ and the predicted flows $\hat{T}_{ij}^{(b)}$.
3. Compute the bootstrap standard error for any scalar quantity $q(\hat{\Theta})$ as:

$$\widehat{\text{SE}}[q] \;=\; \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\Big(q(\hat{\Theta}^{(b)}) - \bar{q}\Big)^2}, \tag{19}$$

where $\bar{q} = B^{-1}\sum_b q(\hat{\Theta}^{(b)})$.

8.2 Confidence Intervals

Report 90% bootstrap percentile confidence intervals for each estimated flow and for each parameter:

$$\text{CI}_{90\%}\Big(\hat{T}_{ij}\Big) \;=\; \Big[Q_{0.05}\Big(\hat{T}_{ij}^{(1)}, \ldots, \hat{T}_{ij}^{(B)}\Big),\; Q_{0.95}\Big(\hat{T}_{ij}^{(1)}, \ldots, \hat{T}_{ij}^{(B)}\Big)\Big], \tag{20}$$

where Q_p denotes the p-th empirical quantile.

Remark. *If the number of distance bins is very small ($K \leq 3$), the bootstrap distribution of $\hat{\beta}$ and $\hat{\gamma}$ may be wide because the data provide little curvature in the likelihood surface. In this case, supplement the bootstrap with a sensitivity analysis over plausible values of β and γ (see Section 9).*

9 Sensitivity Analysis

Point estimates can be sensitive to the functional form of the deterrence function and to the regularization strength. Report the following sensitivity checks in the manuscript.

9.1 Deterrence Function Comparison

Fit all three nested forms and compare using AIC:

$$\text{AIC} \;=\; -2\,\ell_{\text{dist}}(\hat{\Theta}) \;+\; 2\,p, \tag{21}$$

where p is the number of free parameters (2 for exponential, 2 for power-law, 3 for Tanner).

Table 2: Deterrence function candidates and their parameter counts.

Model	**Form**	**Free decay parameters**	**Note**
Exponential	$\exp(-\beta\, d_{ij})$	1 (β)	Nested in Tanner ($\gamma = 0$)
Power-law	$d_{ij}^{-\gamma}$	1 (γ)	Nested in Tanner ($\beta \to 0$)
Tanner	$d_{ij}^{-\gamma} \exp(-\beta\, d_{ij})$	2 (β, γ)	Most flexible; preferred default

9.2 Regularization Sensitivity

Re-run the optimization with $\lambda \in \{0,\ 0.01,\ 0.1,\ 1,\ 10\}$ and plot the estimated $\hat{\boldsymbol{\alpha}}$ coefficients and the distance bin residuals $\hat{p}_k - b_k$ as a function of λ. Coefficients that remain stable across this range are reliably identified.

9.3 Distance Bin Robustness

Refit the model using only two distance bins (merge the outer bins) and verify that $\hat{\beta}$ and $\hat{\gamma}$ do not change substantially. Large shifts indicate that the model is sensitive to the chosen bin boundaries.

10 Goodness-of-Fit and Model Comparison

10.1 In-Sample Fit

Report the following in-sample statistics.

Distance bin residuals. For each bin k, report the absolute percentage error:

$$\mathrm{APE}_k = \left| \frac{\hat{p}_k - b_k}{b_k} \right| \times 100\%. \tag{22}$$

A well-fitted model should achieve $\mathrm{APE}_k < 5\%$ for all bins.

Trip length distribution. Compare the full model-implied trip length distribution $\{\hat{T}_{ij}\}$ against the observed bin distribution using a chi-squared goodness-of-fit test:

$$\chi^2 = \sum_{k=1}^{K} \frac{(n\, b_k - n\, \hat{p}_k)^2}{n\, \hat{p}_k}. \tag{23}$$

10.2 Spatial Autocorrelation

If any observed OD counts are available, compute Moran's I on the residuals $T_{ij} - \hat{T}_{ij}$. A significant positive value indicates spatially structured misspecification.

10.3 Model Comparison Baselines

Compare against the benchmarks in Table 3 using AIC and, where reference OD data exist, the Common Part of Commuters (CPC) metric:

$$\mathrm{CPC} = \frac{2 \sum_{i,j} \min\left(\hat{T}_{ij},\, T_{ij}^{\mathrm{ref}}\right)}{\sum_{i,j} \hat{T}_{ij} + \sum_{i,j} T_{ij}^{\mathrm{ref}}}. \tag{24}$$

11 Complete Parameter Summary

12 Implementation Checklist

Step 1. Zone preparation. Delineate N origin and M destination zones. Compute the pairwise road-network distance matrix $\mathbf{D} \in \mathbb{R}^{N \times M}$ using OSMnx.

Step 2. OSM feature extraction. Extract per-zone features (POI count, road density, land-use entropy, building footprint ratio, transit stop count, betweenness centrality). Standardize all features to zero mean and unit variance.

Table 3: Benchmark models for comparison.

Model	**Reference**	**Key difference from proposed**
Unconstrained Gravity	Wilson (1971)	No distance-bin calibration; no L2 regularization
Radiation Model	Simini et al. (2012)	Parameter-free; uses population intervening opportunities
Doubly-Constrained Gravity	Wilson (1971)	Constrains destinations too; requires destination totals
Deep Gravity	Simini et al. (2021)	Neural SIM; requires large OD training set
Proposed Model	*This work*	MLE/MAP; distance-bin calibrated; OSM covariates; bootstrap UQ

Table 4: Parameters, estimation approach, and initialization.

Parameter	**Constraint**	**Initialization**	**Role**
β	> 0	$1/\bar{d}_{\text{obs}}$	Exponential distance decay rate
γ	≥ 0	1.0	Power-law short-distance exponent
α_p $(p = 1, \ldots, P)$	$\in \mathbb{R}$	0	Log-linear OSM covariate effect on attraction
λ	> 0	0.1 (or cross-validate)	L2 regularization strength

Step 3. Compute initialization. Compute $\bar{d}_{\text{obs}}$ from Equation (10) and set $\beta^{(0)} = 1/\bar{d}_{\text{obs}}$, $\gamma^{(0)} = 1$, $\boldsymbol{\alpha}^{(0)} = \mathbf{0}$.

Step 4. Select λ. Use 5-fold cross-validation on the distance bin likelihood (Section 7), or set $\lambda = 0.1$ as a default.

Step 5. Optimize. Run L-BFGS-B from at least 10 random restarts (Algorithm 1). Check convergence via gradient norm $< 10^{-5}$ and objective change $< 10^{-8}$.

Step 6. Sensitivity analysis. Compare AIC across the three deterrence function forms (Table 2). Vary λ over the recommended grid (Section 9.2).

Step 7. Bootstrap uncertainty. Run $B = 500$ parametric bootstrap replicates. Compute 90% confidence intervals for all $\hat{T}_{ij}$, $\hat{\beta}$, $\hat{\gamma}$, and $\hat{\boldsymbol{\alpha}}$ (Section 8).

Step 8. Goodness-of-fit. Report APE_k for all bins, the χ^2 statistic, and Moran's I on residuals (Section 10).

Step 9. Model comparison. Fit radiation and unconstrained gravity baselines. Compare AIC and CPC in a summary table (Table 3).

13 Recommended Software Stack

Table 5: Software tools for implementation.

Task	**Tool**	**Language**
Zone delineation & mapping	GeoPandas, QGIS	Python / GUI
Road network & distance matrix	OSMnx	Python
OSM feature extraction	OSMnx, Overpass API	Python
Optimization	`scipy.optimize.minimize` (L-BFGS-B)	Python
Auto-differentiation (optional)	JAX (`jax.grad`)	Python
Bootstrap resampling	NumPy `np.random.multinomial`	Python
Model comparison (AIC/CPC)	Custom; or `statsmodels`	Python
Spatial autocorrelation	PySAL / `esda`	Python
Visualization	Matplotlib, Seaborn	Python

Remark. *For automatic gradient computation, implement the objective in JAX and use* `jax.grad` *to obtain exact gradients without manual derivation. This is especially useful if additional covariate transformations or non-standard penalty terms are introduced.*

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